

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Narrate any fire incidence you know about. Describe the reason of fire and the damaged. How the fire could be prevented or managed?
- Q.24 Describe the emergency evacuation process in case of aircraft fire.
- Q.25 Explain the types of fires, fire extinguishers and building design norms w.r.t. fire safety.

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Roll No.

1st Sem / Fire Tech. & Safety

Subject : Basics of Fire Science

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 Fire can be reduced by removing?
- a) Oxygen b) Fuel
- c) Heat d) Any of the above
- Q.2 What type of extinguisher is used for type B fires?
- a) CO₂ filled b) Water filled
- c) Foam filled d) All of the above
- Q.3 Metal comes under which type of fire?
- a) Class D b) Class A
- c) Class B d) Class C
- Q.4 What is the primary mechanism by which smoke and heat spread to upper floors in a multi-story building during a fire?

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- a) Conduction b) Radiation
- c) Convection d) Direct flame contact

Q.5 Which of the following factors does NOT significantly influence the movement of smoke within a building fire

- a) Weather conditions
- b) Building construction (e.g., tightness of construction, open passages)
- c) Operation of the HVAC system
- d) The color of the smoke detector

Q.6 What action is crucial regarding fire doors during a fire to limit the spread of smoke and fire?

- a) Open them to allow smoke to vent out
- b) Ensure they are kept closed
- c) Prop them open for faster evacuation
- d) Remove them completely

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 What is ABC powder?

Q.8 Which fire comes under B type?

Q.9 What is required for fire to happen?

Q.10 What do you mean by super-cruise?

Q.11 What are the types of combustion?

Q.12 What is NBC?

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 What is fire?

Q.14 Why fire spreads in forests?

Q.15 How many types of fire extinguishers are used in case of fire?

Q.16 Which type of building need escape route?

Q.17 What are the classes of fire?

Q.18 How the electrical equipment fire is managed?

Q.19 What ISI standard is relevant fire?

Q.20 What is total flooding system?

Q.21 How the fire is detected?

Q.22 Describe general fire safety requirements?